

1. Executive Summary & Architectural Overview

The **Aarogyam Homeopathy Platform** is an enterprise digital healthcare portal built for clinical homeopathic consultations, video teleconsultations, Hyderabad in-person home visits, and secure digital medical vault storage. Designed strictly under clinical healthcare visual standards (no neon/glittery aesthetics), it utilizes a **Decoupled Cloud-Synchronized Architecture** pairing a zero-bloat client application with a managed **Supabase PostgreSQL Cloud Backend** with Row Level Security (RLS) data isolation.

FRONTEND LAYER - Semantic HTML5 & Modular Vanilla CSS3 - Zero framework overhead (Instant First Contentful Paint) - Custom 5-layer design token system - WCAG AA+ high medical legibility	BACKEND & CLOUD DB - Supabase Managed PostgreSQL Engine - PostgREST v14.5 Auto-Generated REST API - Granular Row Level Security (RLS) Policies - Realtime WebSocket subscriptions	DUAL ACCESS CONTROL - Patient Portal (Rajesh Kumar P-101) - Doctor Portal (Dr. Harini Srinivasan BHMS) - Digital Rx builder & report vault - Hyderabad home visit validator
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2. Frontend Technology Stack

Layer / Component	Technology & Version	Architectural & Clinical Rationale
Document Structure	HTML5 Semantic Standard	Accessible clinical layout, native form validations, ARIA landmarks, modal dialogs for teleconsultation booking and dual portals.
Styling Architecture	Modular Vanilla CSS3 <i>(Zero external CSS frameworks)</i>	High-performance zero-runtime CSS. Deep clinical palette (Forest Green #0D4A38, Sage #E7F2ED, Medical White). Strict rejection of noisy, neon, or glittery gradients.
CSS Architecture Files	5-Layer Modular Structure: - variables.css - base.css - components.css - portals.css - responsive.css	Clean separation of design tokens, core layout, cards, tabs, patient/doctor dashboards, responsive breakpoints (mobile, tablet, desktop).
Typography	Google Fonts: - Outfit (Headlines) - Plus Jakarta Sans (Body)	Authoritative, premium editorial feeling for clinical trust; crystal-clear legibility for patient instructions, lab results, and dosage charts.
Client-Side Logic & State	Vanilla ES6+ JavaScript (Native Module Pattern)	Modular JS architecture with app.js, consultation.js, portals.js, and data.js. Zero compilation requirement.
Local Persistence Cache	Web Storage API (localStorage) Key: aarogyam_clinic_store_v1	Optimistic UI updates, instant portal switching, and offline resilience if cloud connection is interrupted.
Cloud Client SDK	@supabase/supabase-js v2.49.1 <i>(UMD CDN bundle)</i>	Official Supabase JavaScript client managing authenticated cloud calls, query building, error handling, and live table synchronization.

3. Backend, Database & Cloud Infrastructure

The backend is powered by **Supabase Cloud**, a modern open-source Firebase alternative built natively on PostgreSQL. The project has been initialized, linked, migrated, and synchronized with active clinical records.

Infrastructure Layer	Specification / Parameter	Clinical & Security Role
Cloud Provider	Supabase Cloud (AWS)	Managed PostgreSQL 15+ containerized instance with automated daily snapshots, WAL archiving, and point-in-time recovery.
Project Identifier	eiubfnurikvcgsbgpkeh	Dedicated project reference on Supabase Cloud infrastructure.
Project API Endpoint	https://eiubfnurikvcgsbgpkeh.supabase.co	Primary HTTPS API gateway serving PostgREST queries, auth, and storage assets.
API Layer Engine	PostgREST v14.5	Translates client SDK calls into optimized SQL queries directly within Postgres without intermediate Node.js/Python server bottlenecks.
Data Security Model	PostgreSQL Row Level Security (RLS)	Strict per-row access control policies: Patients can only query and view their personal records; verified doctors can inspect queues and write care plans.
CLI & Migration Tool	Supabase CLI v2.117.0	Version-controlled migration schema under supabase/migrations/ for reproducibility across development and production.
Hosting & Delivery Target	Edge Static Host (Cloudflare Pages / Vercel)	Global CDN edge network with sub-50ms latency, automatic HTTP/3, and SSL/TLS certificate management.

4. PostgreSQL Relational Database Schema

Six core relational tables maintain complete audit trails, clinical charts, and doctor-patient interactions:

Table Name	Key Fields & Types	Clinical Purpose & Relationships
public.doctors	id (TEXT PK), full_name, qualification, registration_no, specialization, experience_years, bio	Verified homeopathic practitioners (e.g. Dr. Harini Srinivasan, BHMS, Reg #TSMC-HOM-2015-8842).
public.patients	id (TEXT PK), full_name, email, phone, age, gender, blood_group, allergies, chronic_conditions	Registered patient clinical profiles with allergy flags and systemic medical history.
public.appointments	id (TEXT PK), patient_id (FK), doctor_id (FK), service_type, date, time, status, locality, chief_complaint	Consultation booking queue supporting Video Telehealth, Phone Call, and Hyderabad Home Visits.
public.patient_reports	id (TEXT PK), patient_id (FK), report_title, report_date, report_type, status, clinical_notes	Diagnostic vault holding CBC labs, allergy panels, endoscopy summaries, and pathology documents.
public.care_plans	id (TEXT PK), appointment_id (FK), patient_id (FK), doctor_id (FK), remedies (JSONB), diet_advice, follow_up	Digital prescriptions containing homeopathic remedies (e.g. Nux Vomica 200CH), potency, dosage, and dietary advice.
public.hyderabad_localities	locality_name (PK), zone, home_visit_available, delivery_time_hours, notes	Hyderabad regional coverage directory: Banjara Hills, Jubilee Hills, Gachibowli, Madhapur, Kukatpally, Secunderabad.

5. Dual Role Portal Functional Architecture

Aarogyam provides distinct role-based views for Patients and Doctors, accessible via direct quick-switch mechanisms without confusing multi-step redirects:

PATIENT PORTAL SPECIFICATION Default User: Rajesh Kumar (Patient ID: P-101) Functional Capabilities: • Active Consultation Status: View upcoming teleconsultations, countdown timers, and join video room button. • Diagnostic Records Vault: Upload laboratory tests (CBC, Thyroid profile, past prescriptions) with doctor review flags. • Digital Care Plan & Rx: Interactive prescription viewer with exact remedy names, potencies (30C/200C), dosages, and 1-click printable view. • Hyderabad Locality Check: Instant verification of doorstep home-visit doctor dispatch across Hyderabad PIN codes. • Medical Profile: Management of known drug allergies and chronic constitutional background.	DOCTOR PORTAL SPECIFICATION Default Physician: Dr. Harini Srinivasan, BHMS (Doc ID: D-201) Functional Capabilities: • Live Patient Consultation Queue: Categorized lists of Today's Patients, Completed Cases, and Pending Home Visits. • Electronic Medical Record (EMR) Inspector: Full patient timeline including past constitutional history, uploaded labs, and progress notes. • Homeopathic Care Plan Builder: Form wizard to prescribe single or dual constitutional remedies, frequency, and dietetic regimens. • Consultation State Controller: Change appointment status from 'Confirmed' to 'In Progress' or 'Completed'. • Dispatch Verifier: Approve or assign home visit consultations based on Hyderabad zone availability.
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6. End-to-End Clinical Data Flow

Step	Action / Trigger	Data Flow, Transformation & Cloud Storage
01	Teleconsultation Booking	Patient fills booking form → Input validated against timing conflicts → Optimistic cache update → Asynchronous write to public.appointments via Supabase REST API.
02	Diagnostic Lab Upload	Patient uploads lab summary → Record appended to local vault → Dispatched to public.patient_reports marked as 'Awaiting Review'.
03	Physician Review & Care Plan	Doctor opens patient chart in Doctor Portal → reviews lab report → issues digital prescription → persisted in public.care_plans with instant status sync.
04	Prescription Access & Dispense	Patient receives real-time update in Patient Portal → downloads or prints high-contrast clinical prescription for pharmacy dispensing.

7. Healthcare Security & Compliance Standards

- **Row-Level Security (RLS):** Ensures patient isolation. No patient can read another patient's medical records or appointments.
- **Telemedicine Practice Guidelines (India 2020):** Mandatory physician council registration numbers displayed on prescriptions, registered patient identity verification, and non-emergency explicit disclaimers.
- **Data Integrity & TLS:** 256-bit SSL/TLS encryption for all in-flight API requests between client browser and Supabase PostgreSQL gateway.
- **High-Contrast Clinical Design:** Full adherence to WCAG 2.1 AA standards; zero strobe, glittery, or fluorescent color grading.